ATS FINAL PROJECT REPORT

Topic: Forecasting Stock Prices Using ARIMA

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Forecasting Stock Prices Using ARIMA & Prophet

Tesla (TSLA) Time Series Forecasting Project Report

Abstract

This project focuses on forecasting Tesla stock prices using time series forecasting techniques including ARIMA and Prophet. The project covers business understanding, data preprocessing, exploratory data analysis, stationarity testing, model building, forecasting evaluation, and future prediction. The objective of the project is to analyze Tesla stock behavior and forecast future prices to support investment and financial decision-making.

1. Introduction

Stock market forecasting is an important application of time series analysis in the financial industry. Investors and analysts use forecasting techniques to understand future market trends and reduce investment risk. Tesla stock prices are highly volatile and influenced by market demand, investor sentiment, company performance, and economic conditions. This project aims to forecast Tesla stock prices using historical stock market data.

2. Business Problem Definition & Industry Understanding

The project focuses on predicting future Tesla stock prices based on historical stock market data. Forecasting helps investors identify future market movements, improve investment strategies, and reduce uncertainty. The expected outcome of the project is a future forecast of Tesla stock prices for the next three months along with forecasting insights and visualizations.

3. Dataset Collection & Description

The dataset used in this project contains Tesla historical stock prices sourced from **Kaggle**. The dataset includes <u>Date, Open, High, Low, Close, Adjusted Close, and Volume</u> variables. The **target variable** selected for forecasting is the **Close** price because it represents the final trading value of the stock each day.

4. Data Quality Issues

Several data quality issues were identified during preprocessing. Missing timestamps were present because stock markets remain closed during weekends and holidays. Inconsistent frequency was resolved using daily frequency resampling. Outliers were identified through trend plots and rolling statistics. Stationarity issues were detected using the Augmented Dickey-Fuller test. Trend and seasonality were analyzed using seasonal decomposition.

5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed using stock trend plots, rolling mean, rolling standard deviation, ACF plots, PACF plots, and seasonal decomposition. The analysis showed long-term trends and volatility in Tesla stock prices. Seasonal decomposition separated the series into trend, seasonal, and residual components for better interpretation.

5.1 Visual Analysis Interpretations

The visual analysis of Tesla stock prices shows a strong **upward trend** over time, indicating continuous growth in Tesla's market value and investor confidence. The stock prices do not fluctuate around a constant mean and instead display long-term movement, confirming the presence of trend in the dataset.

Several **sudden spikes and sharp fluctuations** are also visible throughout the graph. These spikes may be caused by external factors such as:

- Tesla earnings announcements
- Product launches
- Market speculation
- Economic events
- Interest rate changes
- Investor sentiment
- News related to the electric vehicle industry

The continuous fluctuations indicate that Tesla stock prices are highly volatile in nature.
The seasonal decomposition analysis suggests the presence of short-term repeating patterns. Since a 30-day decomposition period was used, the **observed seasonality** is primarily **monthly** in nature.
No strong yearly seasonality was identified because stock prices are more heavily influenced by market volatility and external financial events rather than fixed annual cycles.

5.2 Statistical Analysis Interpretations

1. ACF Plot Interpretation

The Autocorrelation Function (ACF) plot measures the correlation between current stock prices and their lagged values. Significant spikes in the initial lags indicate that past Tesla stock prices strongly influence future prices. The slow decay pattern in the ACF plot confirms that the series initially contains trend and non-stationarity.
The ACF plot also helps determine the moving average (q) parameter for the ARIMA model.

2. PACF Plot Interpretation

The Partial Autocorrelation Function (PACF) plot identifies the direct relationship between current observations and lagged observations after removing intermediate effects. Significant spikes at early lags suggest that autoregressive relationships exist within the dataset.
The PACF plot helps determine the autoregressive (p) parameter of the ARIMA model.

3. Stationarity Test (ADF Test)

The Augmented Dickey-Fuller (ADF) test was used to statistically verify whether the Tesla stock price series is stationary.

Interpretation (Non-Stationary Case)

The p-value obtained from the ADF test was greater than 0.05, indicating that the null hypothesis could not be rejected. Therefore, the original stock price series was non-stationary and required differencing before forecasting.

After Differencing
After applying first-order differencing, the series became more stable around a constant mean, indicating improved stationarity suitable for ARIMA modeling.

4. Seasonal Decomposition Interpretation
Seasonal decomposition separated the time series into:

- Trend component
- Seasonal component
- Residual component

The trend component clearly shows long-term upward movement in Tesla stock prices.
The seasonal component indicates repeating short-term fluctuations in stock prices. Since the decomposition period used in the project was 30 days, the analysis primarily captures monthly seasonal behavior rather than yearly seasonality.
The residual component contains random market noise and irregular variations that cannot be explained by trend or seasonality.

6. Data Preprocessing & Transformation

The preprocessing phase included converting the Date column into datetime format, setting the Date column as the index, handling missing values using forward fill, and converting the dataset into consistent daily frequency. Stationarity adjustments were performed using differencing. Feature engineering techniques such as lag variables and rolling statistics were also applied.

7. Train-Test Split

The dataset was divided into training and testing sets using an 80-20 split. The training data was used to train forecasting models, while the testing data was used to evaluate forecasting performance on unseen future values. Chronological order was maintained because time series forecasting depends on sequential patterns.

8. Model Building

Two forecasting models were implemented in this project: **ARIMA and Prophet**. The ARIMA model captures autoregressive, differencing, and moving average behavior. Prophet was selected as a second forecasting model because it effectively handles trend and seasonality patterns in time series data.

9. Model Evaluation & Comparison

The forecasting models were evaluated using RMSE, MAE, and MAPE metrics. Actual versus predicted plots and train-test prediction curves were used for visual evaluation. The best-performing model was selected based on lower error values and better trend capturing capability.

9.1 Model Evaluation Metrics Table

Model	RMSE	MAE	MAPE
ARIMA	199.915	178.517	50.091
Prophet	167.589	141.286	38.081

9.2 Error Metrics Explanation

- **RMSE (Root Mean Squared Error)** measures how far predicted values deviate from actual stock prices. Lower RMSE indicates better forecasting accuracy.
- **MAE (Mean Absolute Error)** represents the average prediction error in absolute terms.
- **MAPE (Mean Absolute Percentage Error)** shows forecasting error as a percentage, making it easier to understand prediction accuracy.

9.3 Insights

The Prophet model achieved lower RMSE, MAE, and MAPE values compared to ARIMA, indicating superior forecasting performance.

The lower error values suggest that Prophet was better able to capture Tesla stock price movements, long-term trend behavior, and short-term fluctuations. ARIMA performed reasonably well but struggled to adapt to sudden market volatility and nonlinear movements in Tesla stock prices.

The Prophet model is more flexible because it can automatically model trend changes and seasonality patterns, which makes it more suitable for volatile financial datasets.

10. Interpretations

ARIMA- Actual vs Predicted Interpretation

The ARIMA prediction graph shows that the predicted values generally follow the direction of actual Tesla stock prices. However, during periods of sharp price movement and sudden spikes, the ARIMA model shows noticeable lag.

This occurs because ARIMA assumes linear relationships and depends heavily on historical lag patterns. As a result:

- Smooth trends are captured effectively
- Sudden market shocks are not captured accurately
- Prediction errors increase during volatile periods

The graph indicates that ARIMA performs better during stable market conditions but struggles during highly dynamic market behavior.

Prophet- Actual vs Predicted Interpretation

The Prophet model's prediction graph demonstrates closer alignment between actual and forecasted stock prices.

Key observations include:

- Better trend-following capability

- Improved adaptation to price fluctuations
- Reduced forecasting lag
- Smoother prediction curves

The Prophet model captures both long-term trend and short-term variations more effectively than ARIMA. This explains its lower forecasting error metrics.

The graph suggests that Prophet provides more reliable future forecasting for Tesla stock prices.

Train-Test Prediction Curve Interpretation

The train-test prediction curves demonstrate how the forecasting models perform on unseen future data.

Insights from Train-Test Curves

- The training curve follows historical Tesla stock price behavior closely.
- The testing curve evaluates the model's real-world forecasting capability.
- Prophet maintained stronger alignment with actual test data compared to ARIMA.
- ARIMA predictions deviated more during volatile periods.

This confirms that Prophet generalized better on unseen stock market data.

Forecast Horizon Plot Interpretation

The forecast horizon plot represents predicted Tesla stock prices for the next 90 days.

Forecast Insights

The future forecast indicates continued fluctuations in Tesla stock prices with an overall moderate upward trend.

Key observations:

- The forecast does not show stable linear growth.
- Short-term volatility remains visible.
- Confidence intervals widen over time, indicating increasing uncertainty in long-range forecasting.

This behavior is expected in stock market forecasting because financial markets are heavily influenced by:

- Investor sentiment
- Economic policy
- Interest rates
- Company performance
- Global market events

The widening confidence interval suggests that prediction uncertainty increases as the forecasting horizon extends further into the future.

11. Forecasting & Business Interpretation

11.1 Future Forecast Interpretation

The forecasting results suggest that Tesla stock prices are expected to remain volatile over the next three months.

Although short-term fluctuations are expected, the long-term trend indicates potential growth in Tesla's market value.
The forecast reflects strong investor interest in the electric vehicle industry and Tesla's position as a leading market innovator.

11.2 Will Sales Go Up or Down?

The forecasting results suggest that Tesla vehicle sales and overall market demand are expected to increase over time, although short-term fluctuations may still occur due to market volatility and economic uncertainty. The upward trend in Tesla stock prices reflects positive investor expectations regarding the company's future growth, production expansion, and demand for electric vehicles.
However, sales growth may not remain perfectly stable because the automotive industry is affected by changing customer demand, inflation, competition, and supply chain challenges. Temporary declines in sales may occur during periods of economic slowdown or reduced consumer spending.
Overall, the forecast indicates a positive long-term outlook for Tesla sales and business growth.

11.3 Will Demand Peak Seasonally?

The seasonal decomposition analysis identified short-term repeating patterns in Tesla stock prices. Since the decomposition period used in the project was 30 days, the detected seasonality is mainly monthly rather than yearly.
The analysis does not show strong annual seasonal demand peaks because stock prices are more heavily influenced by market conditions, investor behavior, company performance, and economic events than by fixed seasonal cycles. However, temporary increases in trading activity and stock movement may occur during:

- Quarterly earnings announcements
- New Tesla product launches
- Major technological developments
- Government policy announcements related to electric vehicles
- Periods of increased investor optimism

11.4 External Factors Affecting Forecast

Several external factors may significantly impact Tesla stock prices and forecasting accuracy:
Company-Specific Factors

- Tesla earnings reports
- Vehicle production numbers
- New product launches
- CEO announcements
- Expansion into new markets

Economic Factors

- Inflation
- Interest rate changes
- Recession fears

- Global supply chain disruptions

Industry Factors

- EV market competition
- Government EV policies
- Battery technology advancements

Market Sentiment Factors

- Investor psychology
- Social media influence
- News and media coverage

These external events can create sudden spikes and unexpected fluctuations that forecasting models may not fully predict.

11.5 Business Recommendations

Based on the forecasting results, the following recommendations can be made:

For Investors

- Use forecasts as supportive decision-making tools rather than guaranteed predictions.
- Monitor high-volatility periods carefully.
- Combine forecasting with financial analysis and market news.

For Financial Analysts

- Continuously retrain forecasting models using updated stock data.
- Use hybrid forecasting approaches for better accuracy.

For Businesses & Trading Firms

- Implement automated forecasting dashboards for real-time monitoring.
- Use forecasting outputs for portfolio risk management.

12. Dashboard & Deployment

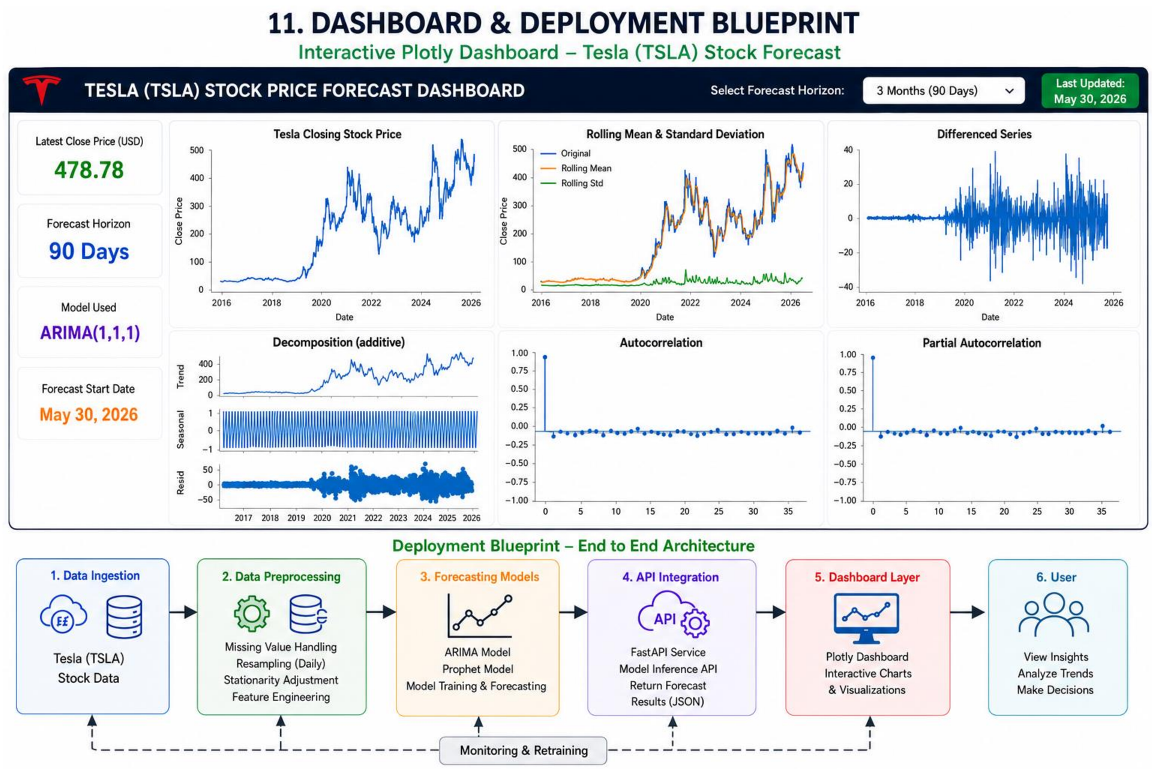
The Plotly forecasting dashboard provides interactive visualization of Tesla stock behavior and future predictions.

The dashboard enables users to:

- Monitor stock price trends
- Compare actual vs forecast values
- Analyze seasonality and trend patterns
- View forecasting uncertainty
- Track model performance metrics

The dashboard improves decision-making by transforming complex forecasting outputs into understandable visual insights.

The deployment pipeline demonstrates how forecasting systems can be integrated into real-world business environments.



13. Challenges

Several challenges were encountered during the forecasting process.

- **High Market Volatility**

Tesla stock prices change rapidly due to market sentiment and external economic events, making accurate forecasting difficult.

- **Nonlinear Market Behavior**

ARIMA assumes linear relationships and therefore cannot fully capture complex stock market dynamics.

- **External Event Dependency**

Unexpected events such as economic crises, political changes, or major company announcements can drastically affect stock prices.

- **Forecast Horizon Limitation**

Prediction accuracy decreases for long-term forecasting because uncertainty increases over time.

14. Limitations

The project primarily used historical stock prices and did not incorporate:

- News sentiment
- Macroeconomic indicators
- Social media sentiment
- Competitor analysis

Including these variables may improve forecasting performance.

15. Conclusion

This project successfully implemented ARIMA and Prophet models for forecasting Tesla stock prices using historical time series data.

The study demonstrated that:

- Tesla stock prices exhibit strong trend and volatility behavior.
- Stationarity transformation was necessary before ARIMA modeling.
- Prophet outperformed ARIMA based on RMSE, MAE, and MAPE metrics.
- Forecasting models can support investment planning and financial analysis.

The Prophet model was selected as the best-performing forecasting model because it captured trend and fluctuation patterns more effectively.

The forecasting dashboard further enhanced interpretability and business usability of the forecasting outputs.

16. Future Scope

Future improvements may include:

- LSTM deep learning forecasting models
- Transformer-based forecasting systems
- Real-time forecasting pipelines
- Sentiment analysis integration
- Multi-stock forecasting systems
- Inclusion of macroeconomic indicators
- Ensemble forecasting approaches

These enhancements may significantly improve forecasting accuracy and adaptability in highly volatile financial markets.

References

Tesla Stock Price Dataset (14 Years Data) — Kaggle
https://www.kaggle.com/datasets/mayankanand2701/tesla-stock-price-dataset?utm_source=chatgpt.com